

《数据库系统原理》实验报告（实验一）					
题目：实验一：DDL 语言实验					
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实验环境：docker OceanBase					
实验步骤及结果截图： 1.连接 Oceanbase <pre>sh-4.4# obclient -uroot@sys -h127.1 -P2881 Welcome to the OceanBase. Commands end with ; or \g. Your OceanBase connection id is 3221225473 Server version: OceanBase_CE 4.2.1.10 (r110000072024112010-28c1343085627e79a4f13c29121646bb889cf901) (Built Nov 20 2024 10:11:18) Copyright (c) 2000, 2018, OceanBase and/or its affiliates. All rights reserved. Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.</pre> 2.创建数据库，命名为 Ecommerce（电商系统），使用数据库 <pre>obclient(root@sys)[(none)]> create database Ecommerce; Query OK, 1 row affected (0.024 sec) obclient(root@sys)[(none)]> use Ecommerce; Database changed obclient(root@sys)[Ecommerce]> █</pre> 3.创建相关数据表，并创建触发器实现 subtotal 为乘积的结果 <pre>obclient(root@sys)[Ecommerce]> create table Customers(-> CustomerID int auto_increment primary key not null, -> Name varchar(50) not null, -> Email varchar(100) not null unique -> check (Email REGEXP '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}\$'), -> Phone varchar(20) not null -> check (Phone REGEXP '^[0-9]{10,15}\$'), -> RegistrationDate date not null ->); Query OK, 0 rows affected (0.088 sec) obclient(root@sys)[Ecommerce]> create table Categories (-> CategoryID int auto_increment primary key not null, -> CategoryName varchar(50) not null unique ->); Query OK, 0 rows affected (0.061 sec) obclient(root@sys)[Ecommerce]> create table Products (-> ProductID int auto_increment primary key not null, -> ProductName varchar(100) not null, -> Price decimal(10,2) not null check (Price > 0), -> StockQuantity int not null check (StockQuantity >= 0), -> CategoryID int not null, -> foreign key (CategoryID) references Categories(CategoryID) ->); Query OK, 0 rows affected (0.066 sec) obclient(root@sys)[Ecommerce]> create table Orders (-> OrderID int auto_increment primary key not null, -> CustomerID int not null, -> OrderDate datetime not null, -> TotalAmount decimal(10,2) not null check (TotalAmount > 0), -> foreign key (CustomerID) references Customers(CustomerID) ->); Query OK, 0 rows affected (0.059 sec)</pre>					

```
obclient(root@sys)[Ecommerce]> create table OrderItems (
->   OrderItemID int auto_increment primary key not null,
->   OrderID int not null,
->   ProductID int not null,
->   Quantity int not null check (Quantity > 0),
->   Subtotal decimal(10,2) not null,
->   foreign key (OrderID) references Orders(OrderID),
->   foreign key (ProductID) references Products(ProductID)
-> );
Query OK, 0 rows affected (0.082 sec)

obclient(root@sys)[Ecommerce]> DELIMITER //
obclient(root@sys)[Ecommerce]> CREATE TRIGGER calculate_subtotal
-> BEFORE INSERT ON OrderItems
-> FOR EACH ROW
-> BEGIN
->   DECLARE product_price DECIMAL(10,2);
->   SELECT Price INTO product_price FROM Products WHERE ProductID = NEW.ProductID;
->   SET NEW.Subtotal = NEW.Quantity * product_price;
-> END //
Query OK, 0 rows affected (0.083 sec)
```

4.表结构查询

```
obclient(root@sys)[Ecommerce]> DESC Customers;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| CustomerID | int(11)    | NO   | PRI | NULL    | auto_increment |
| Name       | varchar(50)| NO   |     | NULL    |              |
| Email      | varchar(100)| NO   | UNI | NULL    |              |
| Phone      | varchar(20)| NO   |     | NULL    |              |
| RegistrationDate | date      | NO   |     | NULL    |              |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.008 sec)

obclient(root@sys)[Ecommerce]> DESC Products;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| ProductID  | int(11)    | NO   | PRI | NULL    | auto_increment |
| ProductName | varchar(100)| NO   |     | NULL    |              |
| Price      | decimal(10,2)| NO   |     | NULL    |              |
| StockQuantity | int(11)    | NO   |     | NULL    |              |
| CategoryID | int(11)    | NO   |     | NULL    |              |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.002 sec)

obclient(root@sys)[Ecommerce]> DESC Categories;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| CategoryID | int(11)    | NO   | PRI | NULL    | auto_increment |
| CategoryName | varchar(50)| NO   | UNI | NULL    |              |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.004 sec)

obclient(root@sys)[Ecommerce]> DESC Orders;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| OrderID    | int(11)    | NO   | PRI | NULL    | auto_increment |
| CustomerID | int(11)    | NO   |     | NULL    |              |
| OrderDate  | datetime   | NO   |     | NULL    |              |
| TotalAmount | decimal(10,2)| NO   |     | NULL    |              |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.005 sec)

obclient(root@sys)[Ecommerce]> DESC OrderItems;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| OrderItemID | int(11)    | NO   | PRI | NULL    | auto_increment |
| OrderID     | int(11)    | NO   |     | NULL    |              |
| ProductID   | int(11)    | NO   |     | NULL    |              |
| Quantity    | int(11)    | NO   |     | NULL    |              |
| Subtotal    | decimal(10,2)| NO   |     | NULL    |              |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.005 sec)
```

5. 查询建立的约束

```
obclient(root@sys)[Ecommerce]> SELECT table_name, constraint_name, constraint_type
-> FROM information_schema.TABLE_CONSTRAINTS
-> WHERE table_name = 'Customers';
```

table_name	constraint_name	constraint_type
Customers	PRIMARY	PRIMARY KEY
Customers	Email	UNIQUE
Customers	Customers_OBCHECK_1744099349772119	CHECK
Customers	Customers_OBCHECK_1744099349772798	CHECK

4 rows in set (0.061 sec)

```
obclient(root@sys)[Ecommerce]> SELECT table_name, constraint_name, constraint_type
-> FROM information_schema.TABLE_CONSTRAINTS
-> WHERE table_name = 'Products';
```

table_name	constraint_name	constraint_type
Products	PRIMARY	PRIMARY KEY
Products	Products_OBCHECK_1744099440024665	CHECK
Products	Products_OBCHECK_1744099440024767	CHECK
Products	Products_OBFK_1744099440025157	FOREIGN KEY

4 rows in set (0.005 sec)

```
obclient(root@sys)[Ecommerce]> SELECT table_name, constraint_name, constraint_type
-> FROM information_schema.TABLE_CONSTRAINTS
-> WHERE table_name = 'Categories';
```

table_name	constraint_name	constraint_type
Categories	PRIMARY	PRIMARY KEY
Categories	CategoryName	UNIQUE

2 rows in set (0.003 sec)

```
obclient(root@sys)[Ecommerce]> SELECT table_name, constraint_name, constraint_type
-> FROM information_schema.TABLE_CONSTRAINTS
-> WHERE table_name = 'Orders';
```

table_name	constraint_name	constraint_type
Orders	PRIMARY	PRIMARY KEY
Orders	Orders_OBCHECK_1744099458022913	CHECK
Orders	Orders_OBFK_1744099458022999	FOREIGN KEY

3 rows in set (0.003 sec)

```
obclient(root@sys)[Ecommerce]> SELECT table_name, constraint_name, constraint_type
-> FROM information_schema.TABLE_CONSTRAINTS
-> WHERE table_name = 'OrderItems';
```

table_name	constraint_name	constraint_type
OrderItems	PRIMARY	PRIMARY KEY
OrderItems	OrderItems_OBCHECK_1744099635465284	CHECK
OrderItems	OrderItems_OBFK_1744099635468213	FOREIGN KEY
OrderItems	OrderItems_OBFK_1744099635465391	FOREIGN KEY

4 rows in set (0.003 sec)

6. 表结构修改与操作

(1) 在 **Customers** 表中添加一个新字段 **EditDate**，类型为 **DATETIME**，当客户每次登录时，该字段更新为某个值（可自行设置，某个固定值即可）。需创建触发器实现此功能。

```
obclient(root@sys)[Ecommerce]> alter table customers
-> add column editdate datetime;
Query OK, 0 rows affected (0.070 sec)

obclient(root@sys)[Ecommerce]> DESC customers;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| CustomerID | int(11)   | NO   | PRI | NULL    | auto_increment |
| Name       | varchar(50) | NO   |     | NULL    |              |
| Email      | varchar(100) | NO   |     | NULL    |              |
| Phone      | varchar(20) | NO   |     | NULL    |              |
| RegistrationDate | date      | NO   |     | NULL    |              |
| editdate   | datetime  | YES  |     | NULL    |              |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.005 sec)

obclient(root@sys)[Ecommerce]> DELIMITER //
obclient(root@sys)[Ecommerce]> create trigger update_customer_editdate
-> after update on customers
-> for each row
-> begin
->     update customers
->     set editdate = current_timestamp
->     where customerid = new.customerid;
-> end;
-> //
Query OK, 0 rows affected (0.054 sec)
```

(2) 将 Customers 中的 name 由 varchar 改为 char(15)。

```
obclient(root@sys)[Ecommerce]> alter table customers
-> modify column name char(15);
Query OK, 0 rows affected (0.038 sec)
```

(3) 修改 Products 表，添加一个 IsFeatured 字段，类型为 BOOLEAN，默认值为 FALSE。同时，创建一个索引，用于快速查询特色商品（IsFeatured 为 TRUE 的商品）。

```
obclient(root@sys)[Ecommerce]> alter table products add column IsFeatured boolean default false;
Query OK, 0 rows affected (0.034 sec)

obclient(root@sys)[Ecommerce]> create index index_of_products_isfeatured on products(isfeatured);
Query OK, 0 rows affected (0.357 sec)
```

出现的问题：

1. 查询建立的约束时，写成

```
SELECT table_name, constraint_name, constraint_type
FROM information_schema.TABLE_CONSTRAINTS
WHERE table_name = Customers;
```

报错

2. 创建触发器时，begin 和 end 之间的 where 语句加上分号会被直接执行

解决方案：

1. 原因：在 SQL 中，不加引号的标识符会被解释为列名

改成

```
SELECT table_name, constraint_name, constraint_type
FROM information_schema.TABLE_CONSTRAINTS
WHERE table_name = 'Customers';
```

2. 在创建触发器的前后使用 delimiter 临时更改 SQL 语句的分隔符为//，从而实现将多条语句看做单条复合语句。另外的解决办法（教科书中写的）：使用 begin atomic-end 语句，同样可以实现将多条语句看做单条复合语句。